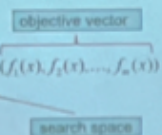


Important Deadlines

Submission deadline	Assessment task/activity
Friday, 7 November 2025 (Week 10)	Submission of a summary for one of the required seminars/lectures you attend
Friday, 7 November 2025 (Week 10)	Submission of presentation materials
Friday, 21 & 28 November 2025 (Week 12 & 13) Saturday, 22 & 29 November 2025 (Week 12 & 13)	Student presentation

Multiobjective Optimization Problem (MOP)

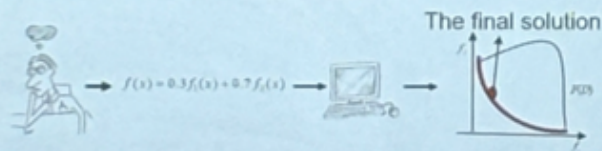
$$\begin{array}{ll} \min & F(x) = (f_1(x), f_2(x), \dots, f_n(x)) \\ \text{subject to} & x \in D \end{array}$$



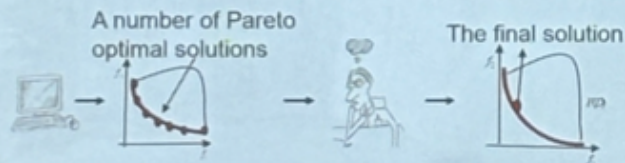
where x can be discrete, continuous or mixed variable.

Basic Strategies in MCDM

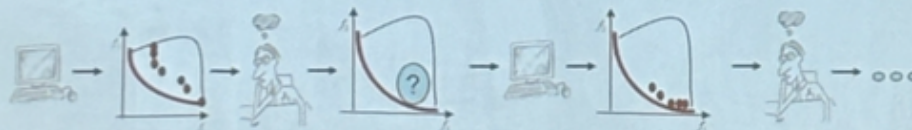
□ A-priori



□ A-posteriori



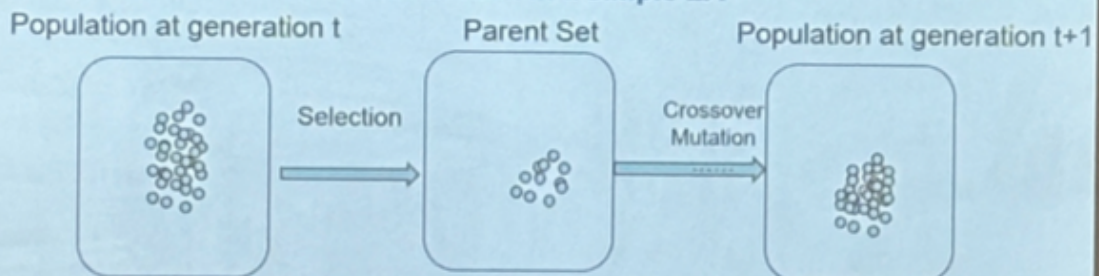
□ interactive



9

Evolutionary Algorithm (EA) =Population Based Iterative Search Method

One iteration in a simple EA



- o Population: a set of candidate solutions.
- o Selection: select fittest solutions to be parents for next generation.
- o Crossover: mix two parent solutions to produce new solutions.
- o Mutation: modify a parent solution to produce a new solution.

Some modern EAs also maintain models and tabu memory

Advantages: 1) Easy to use, a solution can be in any data structure 2) Suitable for black (grey)-box problems.

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